

TickTick:

SSE & TRIZ System Improvement

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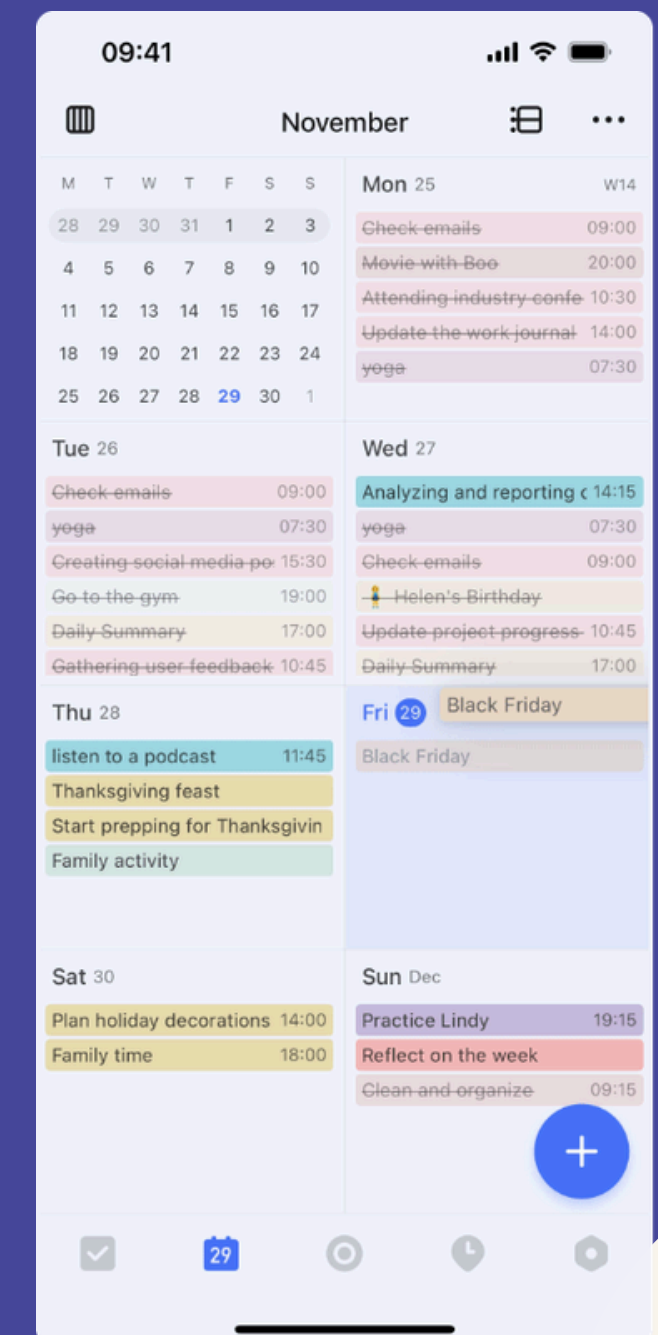
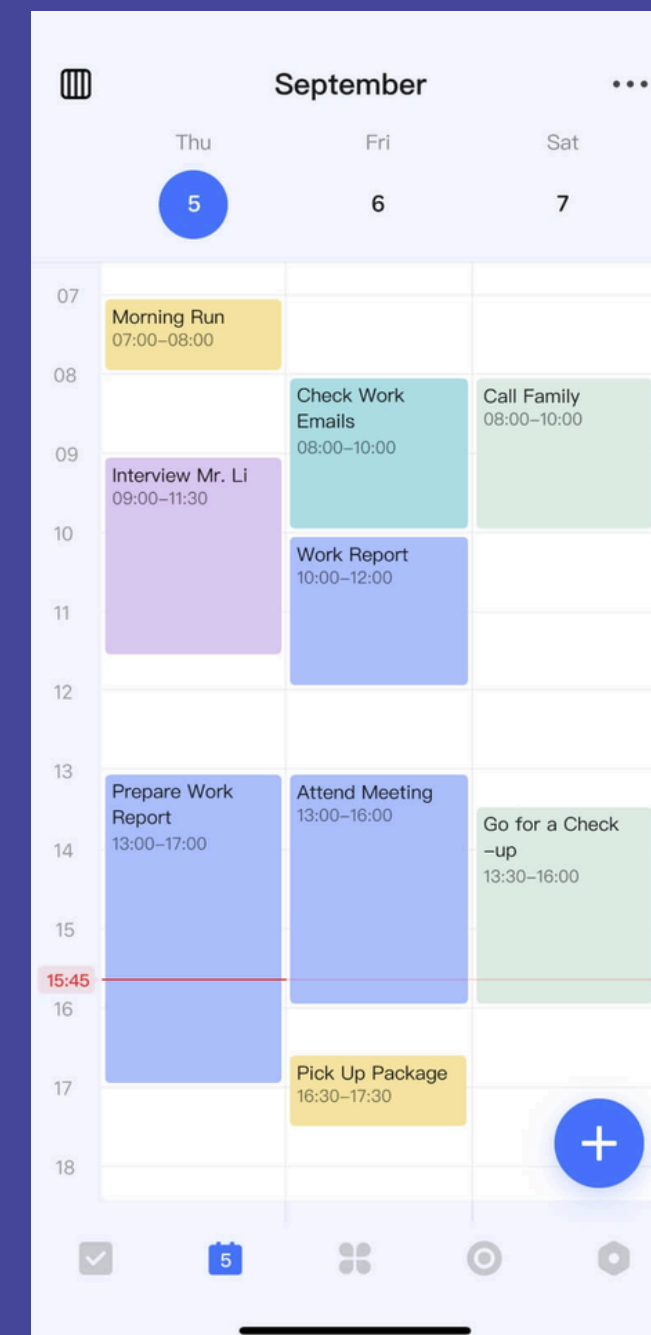
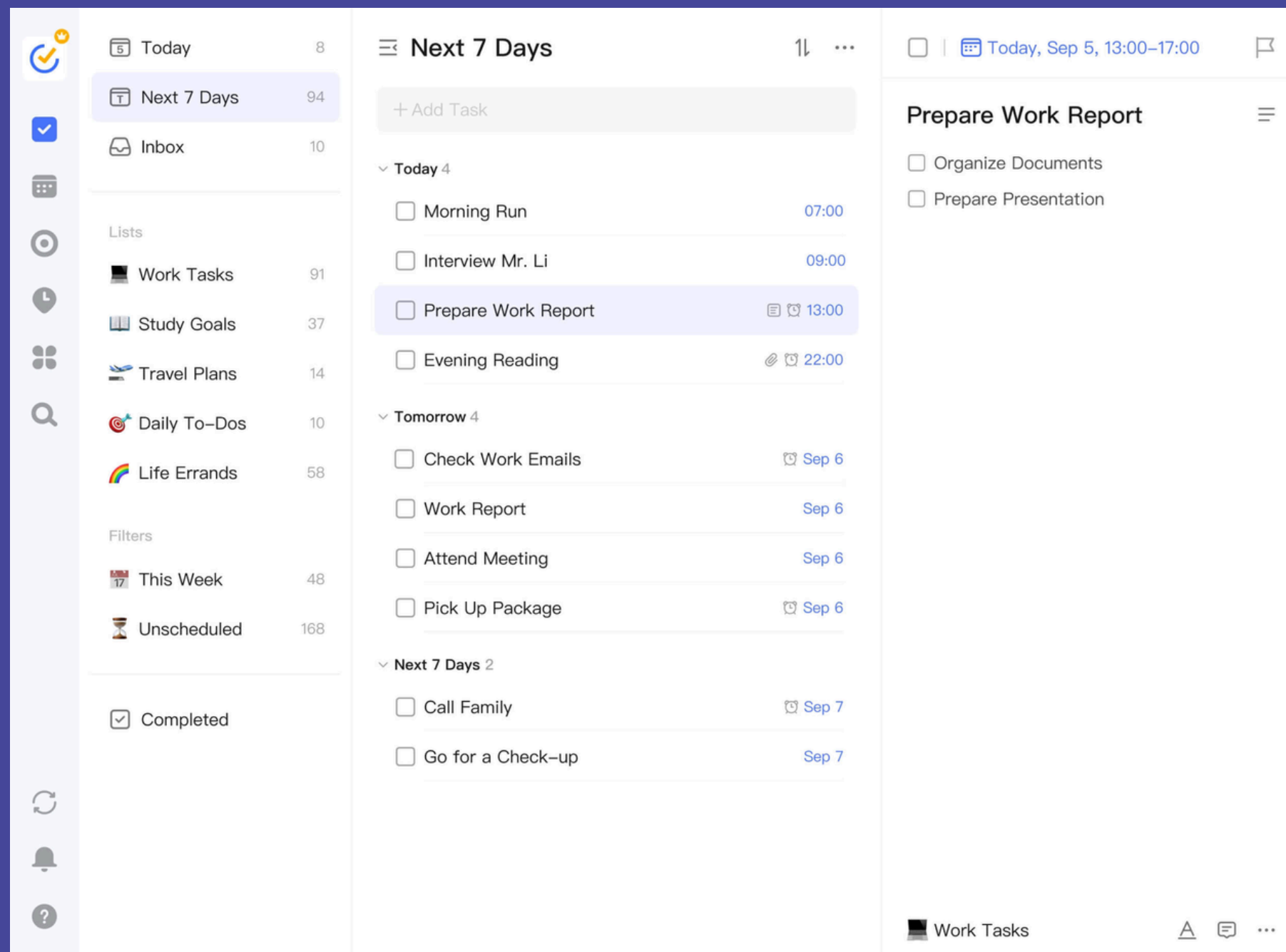
SSE Course Project



SYSTEM SELECTION



TickTick - a cross-platform task and time management system that helps users organize tasks, plan time, and maintain focus.



WHY THIS SYSTEM

- **Widely used real-world productivity application**
- **Clear System Boundaries**
- **Multi-Level Structure**
- **Contains identifiable system-level contradictions**

User

Knowledge worker or student

Task

Manage tasks, deadlines, and long-term goals

Pains:

- Feeling overwhelmed by a large number of tasks
- Forgetting deadlines or important commitments
- Difficulty deciding what to work on next
- Frequent distractions and interruptions
- Lack of motivation to start or complete tasks
- Fragmentation across multiple productivity tools
- Mental fatigue from constant planning and remembering

Gains:

- Clear overview of tasks and priorities
- Reduced mental load through externalized planning
- Improved focus and time awareness
- A sense of accomplishment and visible progress
- Better balance between short-term tasks and long-term goals
- Consistent habit formation and self-discipline

USER VALUE PROPOSITION

SYSTEM OPERATOR MODEL (NINE-WINDOWS)

	Past	Present	Future
Sub-System	Basic task list	Tasks, calendar, Pomodoro, habits	AI prioritization, adaptive reminders
System	Digital to-do list	Integrated productivity platform	Intelligent productivity assistant
Super-System	Simple digital environment	High-distraction digital ecosystem	AI-driven personal support ecosystem

FUNCTIONAL MODEL - CURRENT SYSTEM

System Function	Negative Effect / Bottleneck	Effect on the System
Capture tasks	Complex input (many fields, options)	User friction reduces task capture rate
Structure & prioritize tasks	Too many priorities, tags, and views	Decision fatigue and cognitive overload
Plan tasks in time	Excessive planning effort	Delayed execution, reduced productivity
Remind user	Frequent notifications	Interrupts focus and deep work
Support execution (Pomodoro)	Rigid time blocks	Reduced flexibility for creative tasks
Track progress	Overemphasis on metrics	Stress and pressure instead of motivation

FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

Functional Requirements:

- FR1.** Task Capture
- FR2.** Task Structuring and Prioritization
- FR3.** Time Planning
- FR4.** Reminder and Notification Management
- FR5.** Execution Support
- FR6.** Progress Tracking
- FR7.** Cross-Device Synchronization

Non-Functional Requirements:

- NFR1.** Simplicity of Task Input
- NFR2.** Cognitive Load Control
- NFR3.** Learnability
- NFR4.** Non-Interruptive Notifications
- NFR5.** Flexible Execution Support
- NFR6.** System Responsiveness
- NFR7.** Data Reliability
- NFR8.** Stress-Aware Feedback
- NFR9.** User Control

CONTRADICTIONS IN THE REQUIREMENTS

FEATURE RICHNESS VS. COGNITIVE LOAD

IF the system provides many task attributes, views, and configuration options,
THEN (+) it satisfies functional requirements for flexibility and advanced task management,
BUT (–) it increases cognitive load and decision fatigue.

REMINDERS VS. FOCUS PRESERVATION

IF the system sends frequent notifications and reminders,
THEN (+) it improves reliability and prevents missed deadlines,
BUT (–) it interrupts deep work and reduces user focus.

STRUCTURED EXECUTION VS. USER FLEXIBILITY

IF the system enforces structured focus sessions (e.g., fixed Pomodoro cycles),
THEN (+) it supports sustained concentration and task execution,
BUT (–) it reduces flexibility and may increase stress for users with non-linear or creative workflows.

CONTRADICTION IN PROPERTY

The interruptiveness of the notification system must be high to ensure the user notices tasks and deadlines,
AND simultaneously must be low to preserve user focus and avoid interruptions during deep work.

IMPROVING:

RELIABILITY



WORSENING:

**FOCUS
&
CONCENTRATION**

SOLVING THE CONTRADICTION USING TRIZ TOOLS

TRIZ SOLUTION: CONTEXT-AWARE ADAPTIVE NOTIFICATION SYSTEM

PRINCIPLE 15 - DYNAMIZATION

The notification system dynamically adjusts its behavior based on the user's current context.

PRINCIPLE 10 - PRELIMINARY ACTION

The system anticipates user focus and task urgency to prepare notification behavior in advance.

PRINCIPLE 1 - SEGMENTATION

Notifications are divided into interruptive and non-interruptive modes.

SEPARATION IN CONDITION

The same notification system changes its interruptiveness depending on the user's focus state.

SEPARATION IN TIME

Notifications are delayed and delivered at appropriate moments rather than immediately.

TASK CREATED



**TASK & CONTEXT ANALYSIS
(TASK URGENCY, FOCUS STATE)**



CONTEXT CLASSIFICATION



FOCUS MODE DETECTED
NON-INTERRUPTIVE NOTIFICATION | SILENT /
VISUAL / QUEUED | FOCUS PRESERVED



TASK EXECUTED



NO FOCUS MODE DETECTED
INTERRUPTIVE NOTIFICATION | SOUND / POP-
UP | TASK AWARENESS ENSURED



TASK EXECUTED

FUNCTIONAL MODEL UPDATE

Capture tasks

Structure and prioritize tasks

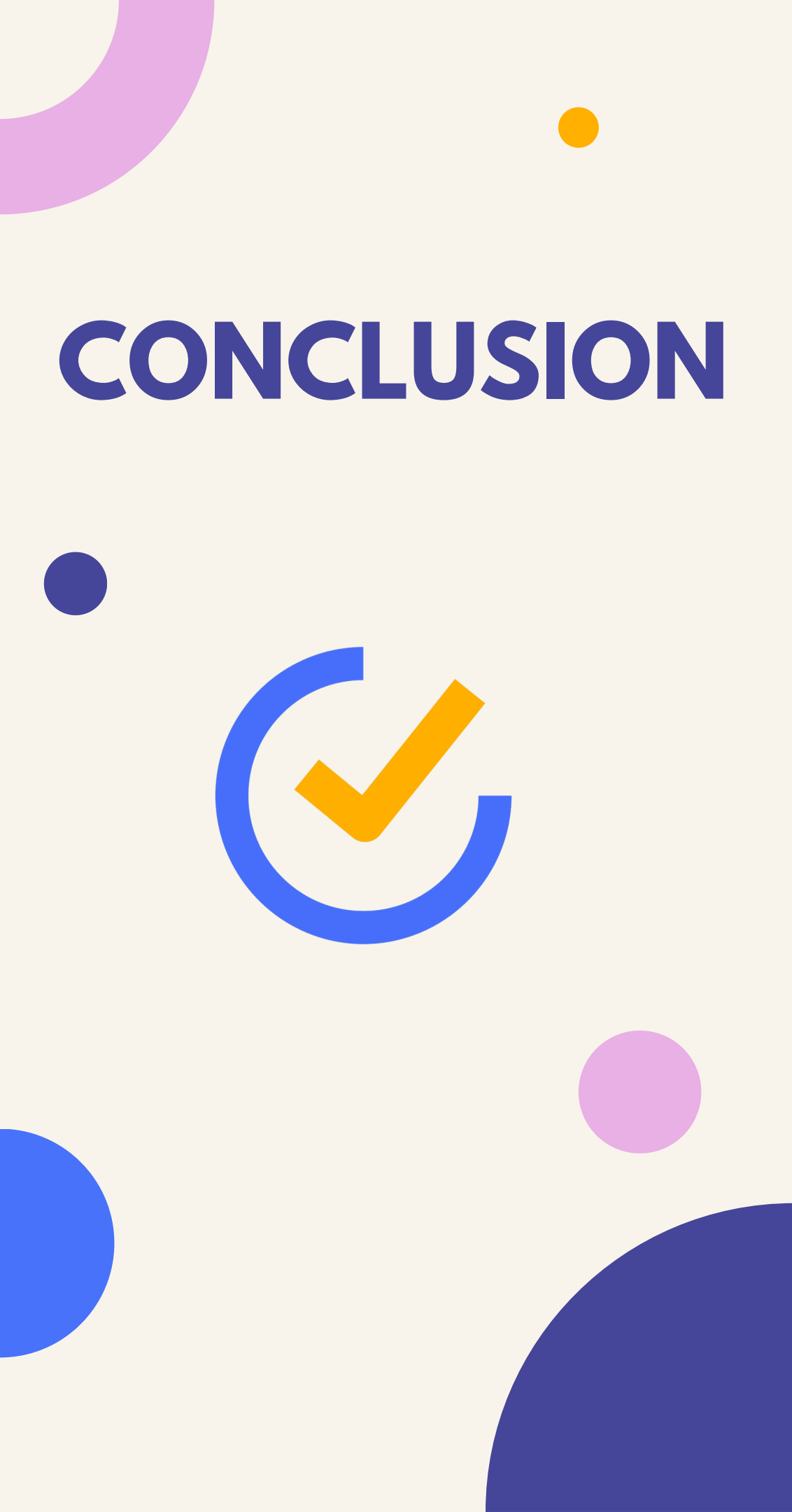
Plan tasks in time

Detect user context (focus / non-focus) ← *new function*

Deliver adaptive notification ← *modified function*

Support execution

Track progress



CONCLUSION



PROBLEM SOLVED

Resolved the contradiction between task awareness and focus preservation by introducing context-aware adaptive notifications.

SYSTEM IMPROVEMENTS

- Preserved user focus without reducing task reliability.
- Reduced cognitive overload caused by interruptive notifications.
- Improved overall productivity through adaptive, context-sensitive behavior.

TRIZ & SSE Application

- Applied TRIZ **Separation in Condition and Time** to resolve a physical contradiction.
- Used **Dynamization, Segmentation, and Preliminary Action** principles.
- Updated the functional model to eliminate a key negative function.

KEY TAKEAWAYS / LESSONS LEARNED

- System contradictions cannot be solved by simple optimization or trade-offs.
- Context-aware, dynamic solutions outperform static “one-size-fits-all” designs.
- Applying SSE and TRIZ enables structured, inventive system improvement aligned with user value.